

Network Detection & Response (NDR) Architecture & Technical Specification



1. Executive Summary & Design Rationale

Modern enterprise networks face sophisticated threat actors employing living-off-the-land techniques, polymorphic command-and-control (C2) channels, and zero-day protocol tunneling. Traditional signature-only Intrusion Detection Systems (e.g., legacy Snort/Suricata without behavioral layers) fail against evasive variants, while standalone machine learning models suffer from catastrophic false-positive fatigue.

This platform implements a **fail-secure, dual-path architecture**:

- **Fast Signature Path**: Suricata rule engine executing against Layer 4-7 protocol payloads for known CVEs and high-rate anomalies.
- **Analytical Behavioral Path**: Supervised XGBoost multi-class classifier + Unsupervised PyTorch Autoencoder calibrated strictly on benign baseline network telemetry.

- **Fail-Secure Arbiter**: A hard defensive guarantee that any component timeout, memory fault, or unhandled exception defaults to immediate containment (`BLOCK_AND_ISOLATE` with `fail_secure=True`)—**never a silent pass**.

2. End-to-End Pipeline Architecture

```
flowchart TD
    subgraph SENSORS ["Sensor & Ingestion Layer"]
        TAP[SPAN / Mirror TAP Interface] --> ZEEK[Zeek Network Security Monitor]
        TAP --> SURI[Suricata IDS Engine]
        ZEEK -->|conn.log, dns.log, ssl.log, http.log| ZPARSER[ZeekParser TSV/JSON Engine]
        SURI -->|eve.json alerts| SPARSER[SuricataParser Engine]
    end

    subgraph FEAT ["Feature Extraction & Preprocessing"]
        ZPARSER --> EXT[FlowFeatureExtractor: 31+ Tabular Features]
        EXT --> ENT[Shannon Payload Entropy]
        EXT --> DNS[DNS Subdomain Entropy & Linguistic Features]
        EXT --> PORT[Port Distribution Shannon Entropy]
        EXT --> TIME[Inter-Arrival Timing & Burstiness Stats]
        ENT & DNS & PORT & TIME --> NORM[NormalizedFlow Common Schema]
    end

    subgraph DETECT ["Dual-Path Detection Engine"]
        SPARSER --> SENG[Suricata Engine: ET Rules & SIDs]
        NORM --> CLF[Supervised XGBoost Classifier]
        Precision-First Threshold 0.85
        NORM --> AE[PyTorch Benign Autoencoder]
        Reconstruction Error MSE
        SENG -->|Signature Match + ATT&CK SIDs| SIG_SIGNS[Detection Signals]
        CLF -->|Multi-Class Probabilities| ML_SIGNS[Detection Signals]
        AE -->|Reconstruction Loss vs Threshold| AE_SIGNS[Detection Signals]
    end

    subgraph DECISION ["Fail-Secure Decision Engine"]
        SIG_SIGNS & ML_SIGNS & AE_SIGNS --> ARBITER[Decision Arbiter]
        WATCHDOG[Watchdog Timeout Guard]
        @fail_secure_guard --> ARBITER
        ARBITER --> VERDICT{Verdict Selection}
        VERDICT -->|Critical Threat / Conf >= 0.85| BLOCK[BLOCK_AND_ISOLATE]
        VERDICT -->|Suspicious / Unseen Channel| QUARANTINE[QUARANTINE_VLAN 99]
        VERDICT -->|Low Risk| PASS[ALERT_ONLY / PASS]
        VERDICT -->|Component Failure / Crash| BLOCK
    end

    subgraph RESPONSE ["Automated Containment & SIEM"]
        BLOCK & QUARANTINE --> MGR[ContainmentManager]
        MGR -->|Primary REST API| OPNSENSE[OPNsense Firewall Gateway]
        MGR -->|Failover Fallback| IPTABLES[Local Host iptables Drop]
        ARBITER --> ECS[SIEM Dispatcher: ECS JSON Format]
        ECS --> WAZUH[Wazuh / ELK SOC Dashboard]
    end
```

3. Threat Detection Matrix & ATT&CK Alignment

MITRE ATT&CK ID	Technique Name	Primary Detection Path	Secondary Safety Net	Containment Verdict
T1046	Network Service Discovery	Suricata SID 3000001 (SYN Burst)	XGBoost PORT_SCAN Classifier	BLOCK_AND_ISOLATE
T1498.001	Direct Network Flood / DoS	Suricata SID 3000002 (SYN Flood)	XGBoost DDOS_FLOOD Classifier	BLOCK_AND_ISOLATE
T1071.001	Web Protocols C2 Beaconing	XGBoost C2_BEACONING	PyTorch Autoencoder MSE	QUARANTINE_VLAN
T1071.004	DNS Tunneling & Exfiltration	Shannon DNS Subdomain Entropy	Suricata SID 3000004	BLOCK_AND_ISOLATE
T1572	Protocol Tunneling / Evasion	PyTorch Benign Autoencoder (Zero-Day)	Suricata SID 3000005	BLOCK_AND_ISOLATE
T1110.001	Password Guessing / Brute Force	Suricata SID 3000006	XGBoost BRUTE_FORCE	BLOCK_AND_ISOLATE

4. Empirical Performance Guarantee

Evaluated against a **20% Stratified Holdout Split (20,002 test flows)** from the **CICIDS2017 dataset**:

- XGBoost Classifier**: Precision: **0.9998** | Recall: **0.9995** | F1: **0.9996** | FPR: **0.0004**
- PyTorch Benign Autoencoder**: Benign Specificity: **98.84%** (7,673 True Negatives / 90 False Positives out of 7,763 benign holdouts)
- Fail-Secure Latency**: Arbiter verdict resolved in **< 2.5ms** per flow.